

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                               |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 4a | <p>The heat input is used to <u>break intermolecular bonds between water molecules/ increasing the potential energy of molecules</u>, and <u>do work against the atmosphere</u> as it expands when it changes phase.</p> <p>The <u>average kinetic energy of molecules remains unchanged</u>, and hence <u>no change in temperature</u>.</p>                                                                                                                                                                                                                                                 | <p>B1</p> <p>B1</p> <p>B1</p> |
| 4b | <p>As light passes two slits instead of one, the <u>total power that passes through the slits is doubled</u>.</p> <p>(Intensity of central bright fringe increases by four times due to constructive interference as waves from both slits arrive in phase.)</p> <p>At dark fringes, <u>destructive interference</u> as waves from both slits arrive with a phase difference of <math>180^\circ</math>, so the <u>intensity of dark fringes becomes zero</u>.</p> <p>The <u>total power / average intensity delivered onto the screen is hence doubled</u>, so that energy is conserved.</p> | <p>B1</p> <p>B1</p> <p>B1</p> |

|    |                                                                                    |    |
|----|------------------------------------------------------------------------------------|----|
| 5a | weight provides the centripetal force (or acceleration of free fall is centripetal | B1 |
|----|------------------------------------------------------------------------------------|----|